JanGyan — Learn Without Limits

Category 4: Health & Emergency Awareness

Module 4 - Disaster & Emergency Preparedness

Module Overview

From earthquakes and floods to fires and heatwaves, India's geography exposes its citizens to nearly every major natural hazard. This handbook is Module 4 of JanGyan's Health & Emergency Awareness category, and it teaches every citizen how to prepare for, survive, and recover from the disasters most likely to affect Indian homes, schools, and workplaces.

By the end of this handbook, you will know exactly what to do before, during, and after an earthquake, flood, fire, cyclone, heatwave, or lightning storm, how to evacuate a building safely, how to build a family emergency plan and survival kit, and which numbers to call when disaster strikes. This module is designed as a complete, self-sufficient guide.

Learning Objectives

After completing this module, you will be able to:

- Explain the four phases of the disaster management cycle.
- Follow the correct before/during/after actions for earthquakes, floods, fires, cyclones, heatwaves, and lightning.
- Evacuate a building safely and calmly.
- Create a family emergency plan that works for your household.
- Assemble a practical emergency survival kit.
- Recall India's key disaster helpline numbers from memory.
- Separate common disaster-response myths from safe, accurate practice.

Why This Topic Matters

Important

India's National Disaster Management Authority (NDMA) estimates that heatwaves alone caused over 24,000 deaths between 1992 and 2015, while lightning kills over 1,600 people across the country in a typical year. Almost every one of these disaster types has well-tested, simple safety actions that dramatically reduce harm — yet most citizens have never been taught them.

India is prone to a wide range of natural hazards: a large share of the country lies in high earthquake-risk seismic zones, coastal states face regular cyclones, many regions face annual floods, and summer heatwaves have become more frequent and severe. Disaster preparedness is not about living in fear — it is about knowing simple, specific actions in advance, so that if the moment comes, you respond automatically instead of freezing.

This is exactly the gap JanGyan's Health & Emergency Awareness category exists to close.

Introduction

A disaster is a sudden event that seriously disrupts normal life, causing widespread damage, injury, or loss that

overwhelms the affected community's own capacity to cope. Disasters can be natural (earthquakes, floods, cyclones) or human-caused (fires, industrial accidents), and preparing for them is a shared responsibility between government agencies and individual citizens.

Definition

Disaster Preparedness is the knowledge and capacity developed by individuals, communities, and organisations to effectively anticipate, respond to, and recover from the impact of likely, imminent, or current hazards.

In India, the **National Disaster Management Authority (NDMA)** — set up under the Disaster Management Act, 2005 — leads this effort at the national level, supported by State Disaster Management Authorities (SDMAs) and the National Disaster Response Force (NDRF). But the first responder in almost every disaster is never the government — it is the person already standing there.

Fundamental Concepts

1. Types of Disasters

Category	Examples
Geological	Earthquakes, landslides
Hydro-meteorological	Floods, cyclones, heatwaves, lightning
Human-caused / Technological	Fires, building collapses, industrial accidents

2. The Disaster Management Cycle

Every disaster, regardless of type, is managed through the same four connected phases:

Phase	What It Means
Mitigation	Reducing risk in advance — earthquake-resistant construction, embankments, safety codes
Preparedness	Planning, training, and stocking supplies before disaster strikes
Response	Immediate actions taken during and right after the event to save lives
Recovery	Rebuilding and restoring normal life afterward

Remember

This handbook focuses mainly on Preparedness and Response — the two phases where an individual citizen's actions matter most.

Complete Detailed Explanation

Each hazard below follows the same simple structure: what to do before, during, and after it strikes.

3. Earthquake Safety

A large part of India, including the entire Himalayan belt and the North-East, falls in high-risk seismic zones (Zones

IV and V) under the Bureau of Indian Standards' seismic zoning map.

When	What To Do
Before	Secure heavy furniture to walls; know safe spots in each room; keep emergency supplies ready.
During	**Drop, Cover, and Hold On** — drop to the ground, take cover under sturdy furniture, and hold on until shaking stops. Stay away from windows and heavy objects.
After	Check for injuries and gas leaks; expect aftershocks; evacuate if the building is damaged; avoid using lifts.

4. Flood Safety

When	What To Do
Before	Know your area's flood risk; keep documents in a waterproof pouch; identify higher ground nearby.
During	Move to higher floors or ground immediately; never walk or drive through flowing flood water; avoid contact with electrical equipment.
After	Avoid flood water (risk of contamination and disease); check structural safety before re-entering buildings; boil or purify drinking water.

> **Warning**
> Just 15 cm of fast-moving flood water can knock an adult off their feet, and 60 cm can sweep away a car. Never underestimate flowing water.

5. Fire Safety

When	What To Do
Before	Install smoke detectors; know two exit routes from every room; keep a fire extinguisher accessible.
During	Get low under smoke; feel doors before opening; if clothes catch fire — Stop, Drop, and Roll; never use lifts.
After	Do not re-enter the building until cleared by authorities; seek medical help for any smoke inhalation.

To use a fire extinguisher, remember **PASS**: **P**ull the pin, **A**im at the base of the fire, **S**queeze the handle, **S**weep side to side.

6. Cyclone Safety

The India Meteorological Department (IMD) issues cyclone warnings using a simple colour code — Green (no warning), Yellow (watch), Orange (alert, be prepared), and Red (warning, take action).

When	What To Do
Before	Track IMD warnings; secure loose objects and roofing; stock food, water, and torches; know your nearest cyclone shelter.
During	Stay indoors, away from windows; if evacuation is ordered, leave immediately via the recommended route.
After	Avoid downed power lines and damaged structures; use water only after purification; wait for an official "all clear."

7. Heatwave Safety

Ahmedabad launched South Asia's first city Heat Action Plan in 2013, after a 2010 heatwave caused over 1,300

deaths in a single month — a model NDMA has since encouraged more than 100 Indian cities to adopt.

When	What To Do
Before	Check local heatwave alerts; plan outdoor work for early morning or evening.
During	Stay hydrated; avoid direct sun between 12–3 pm; wear light, loose clothing; use ORS if sweating heavily.
After	Watch for heat exhaustion or heat stroke signs (high body temperature, confusion, dry skin) and seek medical help immediately if they appear.

8. Lightning Safety

India records well over a thousand lightning deaths every year. The India Meteorological Department, with IITM Pune, offers the free **Damini** app, which can alert users up to about 40 minutes before lightning is likely to strike nearby.

When	What To Do
Before	Check forecasts before outdoor work; keep the Damini app installed if you spend time outdoors.
During	Follow the 30-30 rule — if thunder follows lightning within 30 seconds, go indoors, and wait 30 minutes after the last thunder before going back out. Avoid open fields, tall isolated trees, and water bodies.
After	Check on outdoor workers and livestock; seek medical help immediately for anyone struck.

9. Building Evacuation

- Know at least two exit routes from your home, workplace, or school in advance.
- Walk quickly and calmly — never run or push.
- Never use lifts during a fire or earthquake evacuation.
- Assist elderly people, children, and persons with disabilities if it's safe to do so.
- Go directly to the designated assembly point and stay there until accounted for.

10. Family Emergency Plan

1. Choose two meeting points — one near home, one outside the neighbourhood.
2. Designate an out-of-town contact everyone can reach if local networks are down.
3. Keep copies of key documents (ID, insurance, medical records) in a waterproof folder.
4. Make sure every family member knows the plan, including children.
5. Practise the plan once or twice a year.

11. Emergency Survival Kit

Keep a ready "go-bag" that any family member could grab in under a minute:

Category	Items
Water & Food	Bottled water, non-perishable snacks, energy bars
Light & Power	Torch, spare batteries, power bank
First Aid	Basic first aid kit, essential prescribed medicines

Documents & Money	ID copies, insurance papers, some cash
Communication	Charged mobile phone, list of emergency contacts
Personal	Whistle, mask, warm clothing, sturdy footwear

12. Disaster Helpline Numbers

Helpline	Purpose
112	Unified national emergency number (police, fire, medical)
1070 / 1078	NDMA / State disaster management control rooms
108 / 102	Ambulance services
101	Fire services
1800-116-117	National Poisons Information Centre

Important Terminologies

Term	Simple Meaning
Disaster	A sudden event causing serious disruption and loss, beyond a community's normal coping capacity.
Mitigation	Actions taken in advance to reduce future disaster risk.
Seismic Zone	A geographic area classified by its earthquake risk level.
Evacuation	Organised movement of people away from danger to a place of safety.
Assembly Point	A designated safe location where evacuees gather to be accounted for.
Heat Action Plan (HAP)	A city or state-level early warning and response plan for extreme heat.
Go-Bag	A pre-packed emergency kit ready to grab at a moment's notice.

Step-by-Step Understanding

Building Your Household's Preparedness, Step by Step

1. **Assess your risks:** Identify which hazards are most likely in your area (earthquake zone, flood-prone, cyclone coast, etc.).
2. **Make the plan:** Agree on meeting points, an out-of-town contact, and key responsibilities for each family member.
3. **Build the kit:** Assemble a go-bag using the checklist in this handbook.
4. **Practise:** Run a home fire or earthquake drill at least once a year.
5. **Stay informed:** Follow IMD and NDMA alerts, and install relevant apps like Damini.
6. **Review:** Update your plan and kit whenever your household or risks change.

Real-Life Indian Examples

Example 1

During a moderate earthquake felt across Delhi-NCR, an office worker who had practised "Drop, Cover, Hold On" during a building drill calmly guided colleagues under desks instead of rushing for the stairwell — avoiding the far greater risk of a stampede.

Example 2

Ahead of a cyclone warning on the Odisha coast, a coastal family who kept a stocked go-bag and knew their nearest cyclone shelter evacuated early and calmly, well before roads became unsafe.

Example 3

A farmer in rural Bihar, using the Damini lightning alert app, took shelter well before a storm arrived — an increasingly common example of how simple technology is closing an old awareness gap.

Example 4

During a severe heatwave in Ahmedabad, a construction crew shifted its working hours to early morning after seeing an orange heat alert on public display boards, avoiding the most dangerous midday hours entirely.

Practical Scenarios

Scenario	What You Should Do
The ground starts shaking while you're at home	Drop, take cover under sturdy furniture, and hold on until the shaking stops.
Flood water is rising quickly near your street	Move to higher ground immediately; never attempt to walk or drive through moving water.
You smell smoke in an office building	Alert others, use the stairs (never lifts), and go directly to the assembly point.
A red cyclone warning is issued for your coastal town	Follow official evacuation instructions immediately and head to the nearest shelter.
You're outdoors and hear thunder soon after seeing lightning	Go indoors right away and wait at least 30 minutes after the last thunderclap.

Government Rules & Guidelines

Body / Programme	Role
National Disaster Management Authority (NDMA)	India's apex disaster management body, framing national policy and guidelines for every hazard type.
National Disaster Response Force (NDRF)	Specialised force for large-scale disaster response and rescue operations.
State Disaster Management Authorities (SDMAs)	Implement NDMA policy at the state level, tailored to local risks.

India Meteorological Department (IMD)	Issues cyclone, heatwave, and severe weather warnings, and colour-coded alerts.
Indian Institute of Tropical Meteorology (IITM)	Developed the Damini lightning alert app with IMD.

Laws & Policies

The Disaster Management Act, 2005

This Act established the National Disaster Management Authority, chaired by the Prime Minister, along with State and District Disaster Management Authorities. It created a legal framework for disaster mitigation, preparedness, response, and recovery across India, and mandates that every state and district prepare its own disaster management plan.

Building safety codes such as the **National Building Code of India** and the **Bureau of Indian Standards' seismic zoning provisions (IS 1893)** set out structural safety requirements — including earthquake-resistant design — for buildings across the country's different risk zones.

Important Statistics

Statistic	Insight
NDMA estimates over 24,000 heatwave deaths in India between 1992 and 2015	Shows heatwaves are a serious, often underestimated disaster risk.
Over 1,600 lightning deaths recorded across 16 states in a single year (2020–21)	India has among the highest lightning fatality rates in the world.
Ahmedabad's Heat Action Plan is credited with averting an estimated 1,100 deaths a year	Demonstrates how a well-designed early warning system saves lives at city scale.
A large share of India lies in high-risk seismic Zones IV and V	Includes the entire Himalayan belt and North-East, and parts of western India.

Suggested Visual: A map of India's seismic zones alongside a cyclone-prone coastline overlay.

Best Practices

- Know your area's specific hazard risks — don't rely on generic advice alone.
- Keep your emergency kit updated and easily accessible.
- Practise evacuation drills with your whole family at least once a year.
- Follow official IMD/NDMA alerts rather than rumours or social media forwards.
- Teach children simple actions like "Drop, Cover, Hold On" from an early age.
- Keep important documents digitally backed up as well as in physical waterproof storage.

Common Mistakes

- Standing in doorways during an earthquake — modern doorframes offer little extra protection.
- Using lifts during a fire or earthquake evacuation.
- Walking or driving through flood water to "save time."
- Ignoring official cyclone or heatwave alerts because "it wasn't bad last time."
- Sheltering under an isolated tree during a lightning storm.
- Waiting until a disaster is imminent to pack an emergency kit.

Myths vs Facts

Myth	Fact
"Standing in a doorway is the safest place during an earthquake."	Drop, Cover, and Hold On under sturdy furniture is safer in most modern homes.
"Shallow, slow-moving flood water is safe to walk through."	Just 15 cm of fast-moving water can knock an adult off their feet.
"Lightning never strikes the same place twice."	Tall structures can be struck repeatedly — this is simply false.
"You should open all windows during a cyclone to equalise pressure."	This is a myth and can increase the risk of injury from flying debris.
"Heatwaves are just uncomfortable, not genuinely dangerous."	Heatwaves are among India's deadliest disasters, especially for outdoor workers and the elderly.

Do's and Don'ts

Do	Don't
Drop, Cover, and Hold On during an earthquake	Don't run outside during the shaking itself
Move to higher ground during a flood	Don't drive or walk through moving flood water
Use the stairs during a fire evacuation	Don't use lifts under any circumstance
Follow official cyclone/heatwave alerts	Don't rely on rumours or unverified social media posts
Go indoors at the first sound of thunder	Don't shelter under isolated trees or near water

Safety Tips

- Save all key disaster helpline numbers in your phone today, not when disaster strikes.
- Identify your nearest cyclone shelter or earthquake assembly point in advance.
- Install the Damini app if you live in a lightning-prone area or work outdoors.
- Keep at least 3 days' supply of water and non-perishable food at home.
- Check on elderly neighbours and outdoor workers during heatwaves and cyclones.

Emergency

Dial **112** for unified emergency response, or **1070 / 1078** for NDMA and state disaster control rooms.

Practical Checklist

Household Preparedness Checklist

☐ Family emergency plan discussed and understood by everyone

☐ Go-bag packed and stored somewhere accessible

☐ Key disaster helpline numbers saved in every phone

☐ Two evacuation routes known for home, school, and workplace

☐ Heavy furniture secured to walls

☐ Important documents copied and stored safely

☐ At least one earthquake or fire drill practised in the last year

Quick Revision Notes

Revision

- Disaster Management Cycle: Mitigation → Preparedness → Response → Recovery.
- Earthquake: Drop, Cover, Hold On.
- Flood: move to higher ground; never enter moving water.
- Fire: use stairs, never lifts; Stop-Drop-Roll if clothes catch fire.
- Lightning: 30-30 rule; use the Damini app.
- Key numbers: 112 (unified), 1070/1078 (disaster management), 108/102 (ambulance), 101 (fire).

Key Takeaways

Key Takeaway

1. Preparedness turns panic into calm, practised action.
2. Almost every disaster type has simple, specific, well-tested safety steps.
3. A family plan and a ready go-bag cost little but matter enormously.
4. Official IMD/NDMA alerts are always more reliable than rumours.
5. The first responder in most disasters is an ordinary citizen — often you.

Glossary

Disaster — A sudden event causing serious disruption beyond a community's normal coping capacity.

NDMA — National Disaster Management Authority, India's apex disaster policy body.

NDRF — National Disaster Response Force, specialised in large-scale rescue operations.

Seismic Zone — A geographic classification of earthquake risk level.

Heat Action Plan — A city or state early warning and response system for extreme heat.

Go-Bag — A pre-packed kit of emergency essentials, ready to grab instantly.

Assembly Point — A designated safe gathering location after evacuation.

Frequently Asked Questions

1. What does "Drop, Cover, Hold On" mean?

Drop to the ground, take cover under sturdy furniture, and hold on until the shaking stops.

2. Is it safe to stand in a doorway during an earthquake?

Not generally in modern homes — sturdy furniture cover is safer.

3. How much flood water can knock a person over?

As little as 15 cm of fast-moving water.

4. What should I do if my clothes catch fire?

Stop, Drop, and Roll to smother the flames.

5. What does PASS stand for when using a fire extinguisher?

Pull the pin, Aim at the base, Squeeze the handle, Sweep side to side.

6. What colour cyclone warning means "take action now"?

Red — it indicates a warning requiring immediate action.

7. What is the Damini app used for?

It provides real-time and predictive lightning strike alerts across India.

8. What is the 30-30 rule for lightning?

Go indoors if thunder follows lightning within 30 seconds, and wait 30 minutes after the last thunder before going back out.

9. Should I use a lift during a fire?

No — always use the stairs during a fire or earthquake evacuation.

10. What is a Heat Action Plan?

A city or state-level early warning and response system for extreme heat, first developed in Ahmedabad.

11. What should go in a basic emergency survival kit?

Water, non-perishable food, a torch, first aid supplies, documents, cash, and a charged phone.

12. Who established the NDMA?

It was set up under the Disaster Management Act, 2005, and is chaired by the Prime Minister.

13. What number connects to India's disaster management control room?

1070 or 1078, depending on the state.

14. Why shouldn't I drive through flood water?

Moving water as shallow as 60 cm can sweep away a car, and depth can be hard to judge.

15. What are the four phases of the disaster management cycle?

Mitigation, Preparedness, Response, and Recovery.

16. Is it safe to shelter under a tree during lightning?

No — isolated trees are among the most dangerous places to be during lightning.

17. What should a family emergency plan include?

Meeting points, an out-of-town contact, document copies, and a plan every member understands.

18. How often should a household practise an evacuation drill?

At least once a year.

19. What seismic zones cover most high-earthquake-risk parts of India?

Zones IV and V, including the Himalayan belt and North-East.

20. What is the National Disaster Response Force (NDRF)?

A specialised central force trained for large-scale disaster rescue and response operations.

21. Should I open windows during a cyclone to "equalise pressure"?

No — this is a myth and increases the risk of injury from flying debris.

22. What are common heat stroke warning signs?

High body temperature, confusion, dry skin, and rapid pulse — seek medical help immediately.

23. Who should I check on during a heatwave or cyclone?

Elderly neighbours, outdoor workers, and anyone without reliable shelter or resources.

24. What is an assembly point?

A designated safe location where evacuees gather so everyone can be accounted for.

25. Where can I find official disaster alerts for my area?

Through the India Meteorological Department, NDMA, and your State Disaster Management Authority's official channels.

Suggested Infographics, Flowcharts & Tables

- Flowchart: The Disaster Management Cycle as four connected phases.
- Illustration: "Drop, Cover, Hold On" shown in three simple steps.
- Poster: Go-bag contents, labelled and categorised.
- Map: India's seismic zones and cyclone-prone coastline.
- Card: Key disaster helpline numbers for every household to keep visible.

Official References

- National Disaster Management Authority (NDMA), Government of India — guidelines on earthquakes, floods, fire, cyclones, and heatwaves.
- The Disaster Management Act, 2005, Government of India.
- India Meteorological Department (IMD) — cyclone and heatwave warning systems.
- Indian Institute of Tropical Meteorology (IITM), Ministry of Earth Sciences — Damini lightning alert app.
- Bureau of Indian Standards — IS 1893, seismic zoning provisions.
- Ahmedabad Municipal Corporation — Ahmedabad Heat Action Plan.

Official Government Website Links

Resource	Website
National Disaster Management Authority	ndma.gov.in
India Meteorological Department	mausam.imd.gov.in
National Disaster Response Force	ndrf.gov.in

Indian Institute of Tropical Meteorology	tropmet.res.in
Ministry of Home Affairs (Disaster Management Division)	ndmindia.mha.gov.in

Conclusion

Disasters cannot always be prevented, but panic, confusion, and preventable harm often can be. Every safety action in this handbook — Drop, Cover, Hold On; moving to higher ground; using the stairs; going indoors at the first thunderclap — is simple enough to remember under pressure, precisely because it was designed that way by people who study exactly how disasters unfold.

As you move ahead in the JanGyan platform, use the Practice and Quiz sections to reinforce what you've learned here, and don't hesitate to use AI Assistance whenever a doubt arises. Preparedness costs very little today — and can be worth everything on the day it's needed.

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